

PETROLEUM PRODUCTION ENGINEERING

PRE MODULE

COURSE BRIEF

COURSE INTRODUCTION

By

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Course Details

- **Course** **Petroleum Production Engineering**
- **Course Code** **20PEBOE01**
- **Day & Timing** **04:15 AM to 06:10 AM (Monday)**
04:15 PM to 05:10 PM (Tuesday)
- **Credit** **3-0-0 (3 Lectures)**
- **Venue** **E-105**
- **Faculty Details** **Gaurav Hazarika**



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Syllabus

UNIT-1: Petroleum Production Engineering 10 Hrs.

- Role of Production Engineer/activities performed at various levels of field development and its exploitation. Petroleum Production System-Well Head Equipment, Charismas tree, valves, hangers.

UNIT-2: Reservoir Considerations 9 Hrs.

- Flow control devices , Packers , Tubular and Flow Lines, Productivity Index, Inflow Performance, Tubing performance relation, Formation Damage and Well Stimulation.



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Syllabus

UNIT-3: Introduction to Artificial Lift Techniques 10 Hrs.

- Principle and application of artificial lift methods- rod pumps(SRP), Gas Lift(Continuous/Intermittent), Progressive Cavity Pump(PCP), Electric Submersible Pump(ESP), Hydraulic Lifts(Jet Pumps).

UNIT-4: Offshore Production System 10 Hrs.

- Introduction; Offshore Production System-Fixed and Floating Production Platform-Mooring and Anchoring , Deep water Technology, Subsea production system, Flow assurance in deep sea, New Technologies.



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CO's

- CO1:** Understand the role of a production engineer and comprehend the petroleum production system
- CO2:** Understand wellbore performance and well stimulation techniques
- CO3:** Examine artificial lift techniques used in oil and gas industry
- CO4:** Evaluate use of artificial lift techniques used in oil and gas Industry
- CO5:** Analyse various types of shallow and deep water production platforms
- CO6:** Evaluate various offshore production system and deep water technologies

References

- Dr. Guo Boyun, **Computer Aided Petroleum Production Engineering**
- **Deep water Petroleum Exploration & Production** - By William Leffler, Richard Pattardozi, Gordon Sterling
- **Floating Production System** - By N.K.Mitra

PETROLEUM PRODUCTION ENGINEERING

MODULE-1

PETROLEUM PRODUCTION ENGINEERING SURFACE FACILITIES

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Petroleum Production Engineering

- Production engineering is a part of petroleum engineering that attempts to maximize oil and gas production in a cost-effective manner.
- Production engineers need to have a thorough understanding of the petroleum production systems with which they work.

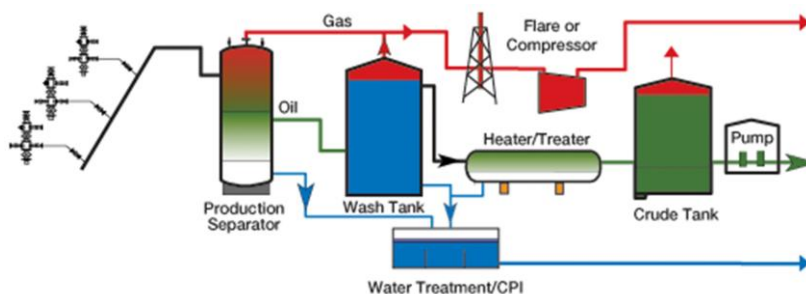


Production System

- The principles of fluid flow through the production system is important in estimating the performance of individual wells and optimizing well and reservoir productivity
- The production system is the system that transports reservoir fluids from the subsurface reservoir to the surface, processes and treats the fluids, and prepares the fluids for storage and transport to a purchaser/sales points/refinery.

Production System

- The basic elements of the production system include the reservoir, wellbore, tubular goods and associated equipment, surface wellhead, flowlines, processing facilities and artificial lift equipment



Production System



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MODULE-2

PETROLEUM PRODUCTION ENGINEERING

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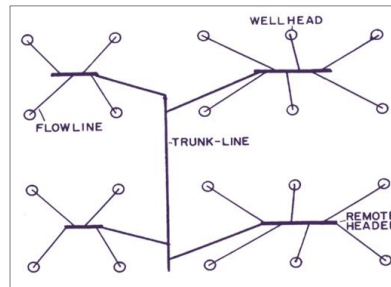
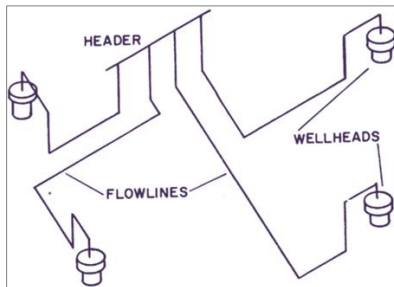
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Surface Gathering System

- The group gathering system consists of flow lines from a well/wells to a centralized *Group Gathering Station*.
- Radial gathering system
- Axial gathering system



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Production System



Well Head

Separator

Header

Heater Treater

Storage Tank

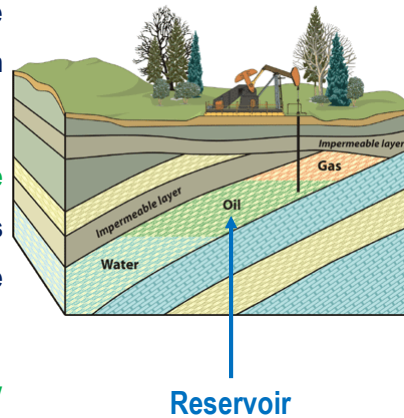


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Production System

- The **reservoir** is the source of crude oil and gas for the production system.
- It is the **porous and permeable** media in which the reservoir fluids are stored and through which the fluids will flow to the wellbore.
- It also furnishes the **primary energy** for the production system.



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MODULE-3

PETROLEUM PRODUCTION ENGINEERING

ROLE OF PETROLEUM PRODUCTION ENGINEERS

By

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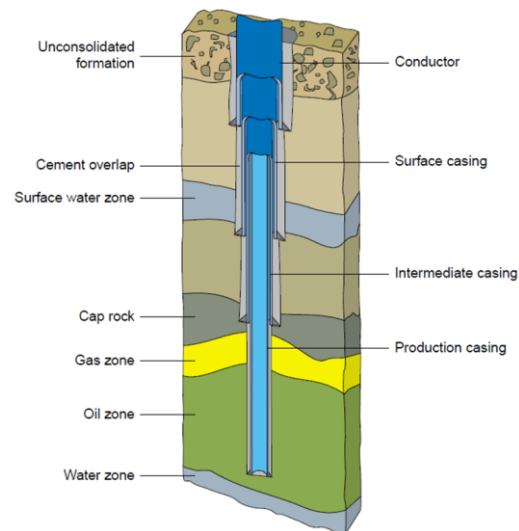
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Role of Production Engineers

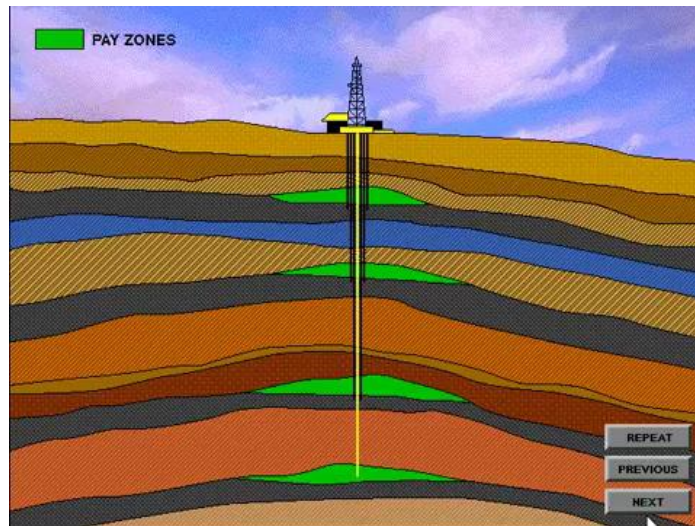
- The role of a production engineer is to **maximize oil and gas production in a cost-effective manner.**
- Familiarization and understanding of oil and gas production systems are essential to the engineers.
- A complete oil or gas production system consists of a reservoir, well, flowline, separators, pumps, and transportation pipelines.

Well to Christmas Tree



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Well to Christmas Tree

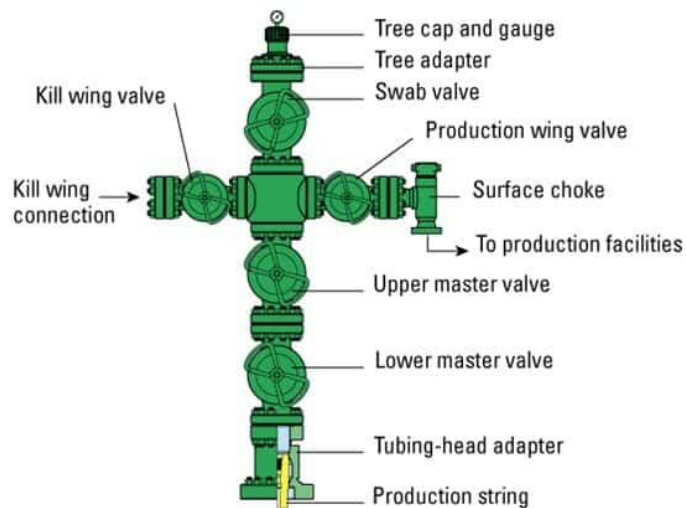


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Activities of Petroleum Production Engineer



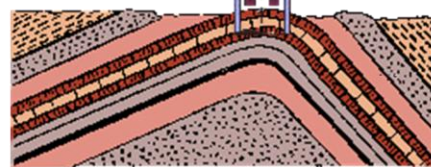
Surface Facilities

Corrosion Control

Artificial Lift



Well Completion,
Testing & Services



Well Stimulation

Water & Gas Control

Sand Control

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Activities of Petroleum Production Engineer

Production Optimization
(Nodal-Analysis)

Production & Injection
Initiation

**Production
Activities**

Other Activities

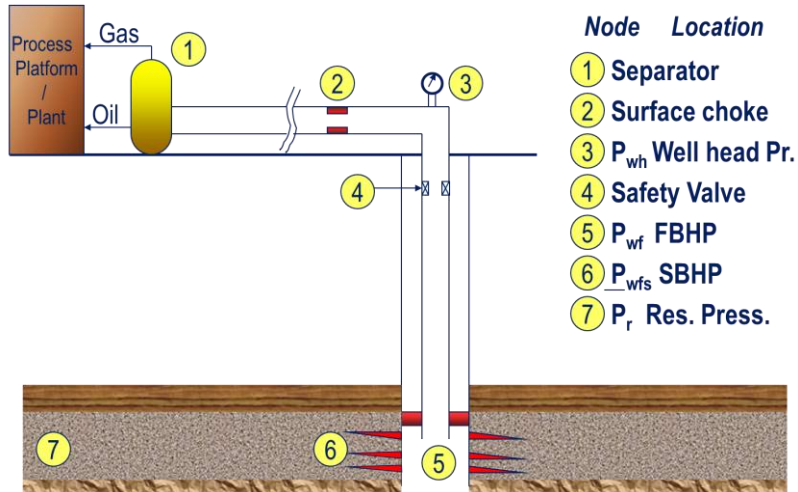
Revival & Enhancement of
Productivity & Injectivity

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Nodal Analysis

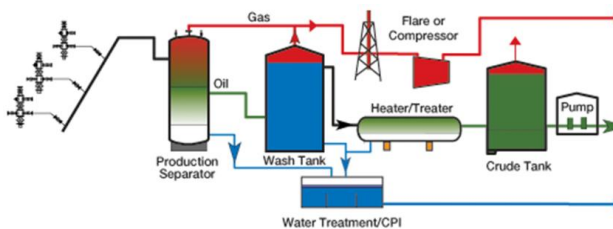


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Production or Injection Initiation

- Well Completion, Testing & Services
- Surface Facilities (Transportation & Processing)
- Pipeline
- Installations
- Platforms
- Plants, etc.
- Continuous Production/Injection Maintenance, Monitoring, etc.



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Development of Surface Facilities

- Depending on the type of crude and its properties a host of other facilities are setup:
 - Heater treaters
 - Bath heaters
 - De-emulsification plants
 - Crude/gas sweetening set up
 - Chemical injection, etc.
- Depending upon the magnitude and scale of operations the various requirements quadruple.

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MODULE-4

PETROLEUM PRODUCTION ENGINEERING

ROLE OF PETROLEUM PRODUCTION ENGINEERS-II

By

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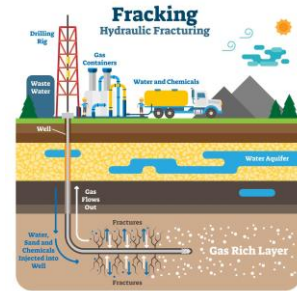
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Revival & Enhancement of Well Productivity/Injectivity

- Well Stimulation
- Water & Gas Control
- Sand Control
- Artificial Lift
- Workover



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Artificial Lift

- In case the reservoir is unable to furnish sufficient energy to produce fluids to the surface, **artificial lift equipment** is used to enhance production rates by adding energy to the production system.
- This component of the system is composed of both surface and subsurface elements.
- This additional energy can be furnished **directly to the fluid** through subsurface pumps, by reducing the backpressure at the reservoir with surface compression equipment to lower wellhead pressure, or by injecting gas into the production string to reduce the flowing gradient of the fluid.

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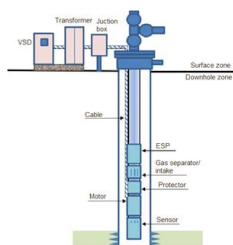
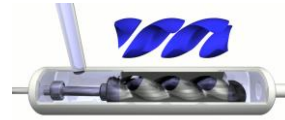
Artificial Lift Photos



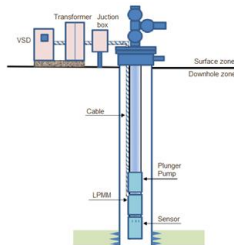
SRP



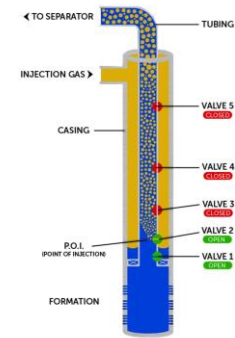
PCP



ESP



Plunger Pump



Gas Lift

Other Activities

- Field / Well Problem Diagnosis
- Corrosion Control
- Field development
- EOR facilities
- Repair & maintenance of flowline
- Plants handling & maintenance
- Well Maintenance Activities



Well Maintenance Activities

- Hot oil circulation
- Scrapping
- Nitrogen & Compressor applications
- Steam application
- Flow line flushing



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Test Your Understanding

- Discuss about the Petroleum Production System from the wells to the Surface Processing Facility/Group Gathering Station (GGS)? **20 Marks**
- Discuss with diagram the different types of gathering used in the Upstream Oil and Gas Industry? **10 Marks**
- Discuss the roles and activities performed by a Petroleum Production Engineer. Analyze and discuss any 5 production activities where the role of a (Chemical/Mechanical/Civil/Computer/Electrical Engineer) plays an integral part in production of Oil and gas. **10+5 Marks**
- Draw and demarcate the well head equipment used in the Upstream Oil and Gas Industry? Also discuss which valves are operated so that the oil flow is directed towards the surface facility **10 Marks**

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